

Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Computer Science research engineer focused on embedded/edge AI systems, heterogeneous runtimes and reliable multi-device execution.

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Canada

Completed Summer 2024

Selected Research and Engineering Projects

Utopia / Digital City — *Multi-device personal-computing and research platform*

2026–present

github.com/zhiheng-zhang-Mera/Utopia

- Built Android and Web control surfaces around a platform-neutral City protocol with authenticated device paths, explicit confirmation and truthful degraded-state handling.
- Current product evidence includes an accepted three-end mesh across two real Windows workers plus Android, with strict target-device routing and cross-host review.
- Research-strengthening stages for experiment registry, provenance/metrics, repeated scenarios, fault injection and replay/ablation have accepted evidence; artifact export remained under independent review at the 2026-10-06 snapshot.
- Planned next direction: a **Personal Compute Fabric** for heterogeneous-device telemetry, placement/offloading, recovery and explainable cross-device execution.

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building long-horizon repository execution with durable task state, restart/recovery, hardware-aware concurrency and explicit terminal-state evidence.
- Uses failure isolation and recovery-oriented execution to study when a software agent has genuinely completed work.
- Supports coding-agent reliability, autonomous maintenance and empirical software-engineering evaluation.

Codex Boss — *Agent orchestration and evidence-based acceptance*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Developed a multi-agent workflow platform for planning, execution, critique and adjudication with durable state and evidence-aware acceptance.
- Treats unresolved failure cases and incomplete acceptance conditions as explicit evidence rather than reporting them as solved.
- Supports trustworthy agent orchestration, completion claims, auditability and human oversight.

Research Interests

Embedded systems; edge AI; low-power wireless; LLM inference systems; multi-agent services; heterogeneous-device scheduling.

Technical and Research Strengths

- End-to-end research engineering across system design, implementation, testing, controlled experiments and evidence-driven iteration.
- Multi-device and long-running software systems with explicit lifecycle, recovery, routing, observability and acceptance boundaries.
- Git/GitHub-based reproducible development, regression testing, cross-host review and structured preservation of failure evidence.

Current Research Direction

I want to study a heterogeneous personal compute fabric that coordinates embedded and general-purpose devices for agent workloads while exposing resource, placement and recovery decisions.